

INTRODUCTION

1.1 Introduction

Inventory and sales management play a crucial role in the success of retail and wholesale businesses. Companies deal with large numbers of products, vendors, and transactions every day. Efficient management of inventory ensures that products are available when customers demand them while avoiding excess stock that increases storage costs and reduces cash flow.

In today's competitive market, organizations must analyze their sales and purchasing data to make better business decisions. By studying patterns in sales, profit margins, vendor contributions, and stock turnover, businesses can identify inefficiencies and opportunities for improvement.

This project focuses on analyzing inventory and sales data to identify factors affecting profitability and operational efficiency. Through data analysis techniques such as summary statistics, correlation analysis, and vendor performance evaluation, the project aims to uncover insights related to product performance, vendor dependency, pricing strategies, and inventory management.

The analysis helps businesses understand which brands need promotional strategies, which vendors contribute the most to revenue, and how bulk purchasing affects unit costs. It also highlights issues such as slow-moving inventory and variations in vendor profitability. These insights support better decision-making and help improve overall business performance.

1.2 Problem Statement

Retail and wholesale companies often face challenges in managing inventory, pricing products correctly, and maintaining profitable vendor relationships. Poor inventory planning may lead to unsold stock, increased storage costs, or lost sales opportunities.

Additionally, companies may become overly dependent on a few vendors, which can create supply chain risks. Some products may have high profit margins but low sales, while others sell in large quantities but generate lower profits.

Without proper analysis of sales and inventory data, it becomes difficult for organizations to identify underperforming brands, optimize purchasing strategies, and maintain balanced vendor relationships. Therefore, there is a need for a data-driven approach to evaluate sales performance, vendor contributions, and inventory turnover in order to improve operational efficiency and profitability.

1.3 Objectives and Goals

The main objectives of this project are:

1. To analyze sales and inventory data to understand overall business performance.
2. To identify underperforming brands that may require promotional or pricing adjustments.
3. To determine the top vendors contributing to total sales and gross profit.
4. To analyze the impact of bulk purchasing on unit costs and profitability.
5. To evaluate inventory turnover in order to reduce holding costs and improve stock management.
6. To compare profit margins between high-performing and low-performing vendors.
7. To provide data-driven recommendations that help improve profitability and operational efficiency.

1.4 Scope of the System

The scope of this project is focused on analyzing sales, purchase, and inventory data to generate meaningful business insights. The system evaluates various factors such as sales revenue, purchase quantities, gross profit, profit margins, and stock turnover.

The analysis helps identify patterns in vendor performance, pricing strategies, and inventory movement. It also highlights slow-moving products, vendor dependencies, and the financial impact of bulk purchasing.

However, the system mainly focuses on **data analysis and decision support**. It does not directly manage real-time inventory transactions or automate purchasing processes. Instead, it provides analytical insights that assist managers in making better strategic decisions related to inventory management, vendor selection, and pricing strategies.

1.5 Existing System and Need of the System

Existing System

In many organizations, inventory and sales data are stored in databases or spreadsheets and are often analyzed manually. Traditional methods rely on simple reports and basic calculations, which may not provide deep insights into business performance.

These systems usually focus only on recording transactions rather than analyzing trends. As a result, companies may struggle to identify slow-moving inventory, understand vendor performance, or detect profitability issues in time.

Manual analysis can also be time-consuming and prone to errors, making it difficult for managers to make quick and informed decisions.

Need of the Proposed System

To overcome these limitations, a data-driven analytical system is required. By applying data analysis techniques and visualization tools, businesses can gain deeper insights into sales trends, vendor contributions, and inventory performance.

The proposed system helps organizations:

- Identify underperforming brands and optimize pricing strategies.
- Reduce vendor dependency and manage supply chain risks.
- Improve inventory turnover and reduce holding costs.
- Understand profitability differences across vendors and products.
- Make informed decisions based on reliable data insights.

Overall, the system supports better inventory planning, cost management, and profit optimization, helping the organization achieve sustainable business growth.

PROPOSED SYSTEM

2.1 Proposed System

The proposed system is a **data analysis–based inventory and sales management system** designed to provide meaningful insights from retail and wholesale business data. The system analyzes sales transactions, purchase records, vendor information, and inventory details to identify patterns that affect profitability and operational efficiency.

The system uses data processing and analytical techniques to evaluate factors such as product sales performance, vendor contributions, profit margins, and inventory turnover. By analyzing these parameters, the system helps in identifying slow-moving products, profitable vendors, and opportunities for cost optimization.

The proposed system also supports decision-making by generating insights about pricing strategies, bulk purchasing advantages, and vendor dependency. Visualization tools and statistical analysis help present the results in a clear and understandable form so that managers can take appropriate actions to improve sales performance and reduce operational risks.

Overall, the system acts as a **decision support tool** that helps businesses optimize inventory management, improve profitability, and enhance vendor relationships.

2.2 Advantages and Disadvantages

Advantages

- 1. Improved Decision Making**
Provides data-driven insights that help managers make better business decisions.
- 2. Efficient Inventory Management**
Helps identify slow-moving and unsold inventory, reducing storage costs.
- 3. Vendor Performance Analysis**
Identifies top-performing and low-performing vendors based on sales and profit contribution.
- 4. Cost Optimization**
Analyzes bulk purchasing strategies to reduce unit costs and increase profit margins.
- 5. Better Profitability Analysis**
Helps identify products and vendors that generate higher profits.
- 6. Time Saving**
Automated analysis reduces the time required for manual data evaluation.

Disadvantages

- 1. Dependence on Data Quality**
The accuracy of results depends on the quality and completeness of the data.
- 2. Initial Setup Effort**
Preparing datasets and configuring analysis tools may require time and technical knowledge.
- 3. Limited to Analytical Insights**
The system provides insights but does not directly control inventory operations.
- 4. Requires Technical Tools**
Users may need knowledge of tools such as Python, Power BI, or Tableau to perform advanced analysis.

2.3 Feasibility Study

A feasibility study evaluates whether the proposed system is practical and beneficial for implementation.

Technical Feasibility

The proposed system is technically feasible because it uses commonly available tools such as **Python, Excel, Power BI, or Tableau** for data analysis and visualization. These tools are widely used and do not require complex infrastructure.

Economic Feasibility

The system is cost-effective because it relies mostly on existing software tools and datasets. The benefits gained from improved inventory management, cost reduction, and better decision-making outweigh the implementation costs.

Operational Feasibility

The system is easy to use and supports business operations by providing clear insights into sales, inventory, and vendor performance. Managers and analysts can easily interpret the results and apply them to improve business strategies.

Time Feasibility

The system can be implemented within a reasonable time frame since it mainly involves data collection, preprocessing, analysis, and visualization. The required technologies and tools are readily available.

2.4 System Requirements

Software Requirements

The following software tools are required to implement the proposed system:

- **Operating System:** Windows 10 or higher
- **Programming Language:** Python
- **Data Analysis Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Data Visualization Tools:** Power BI / Tableau
- **Development Environment:** Jupyter Notebook / VS Code / Anaconda
- **Database or Data Source:** Excel / CSV files / SQL Database

Hardware Requirements

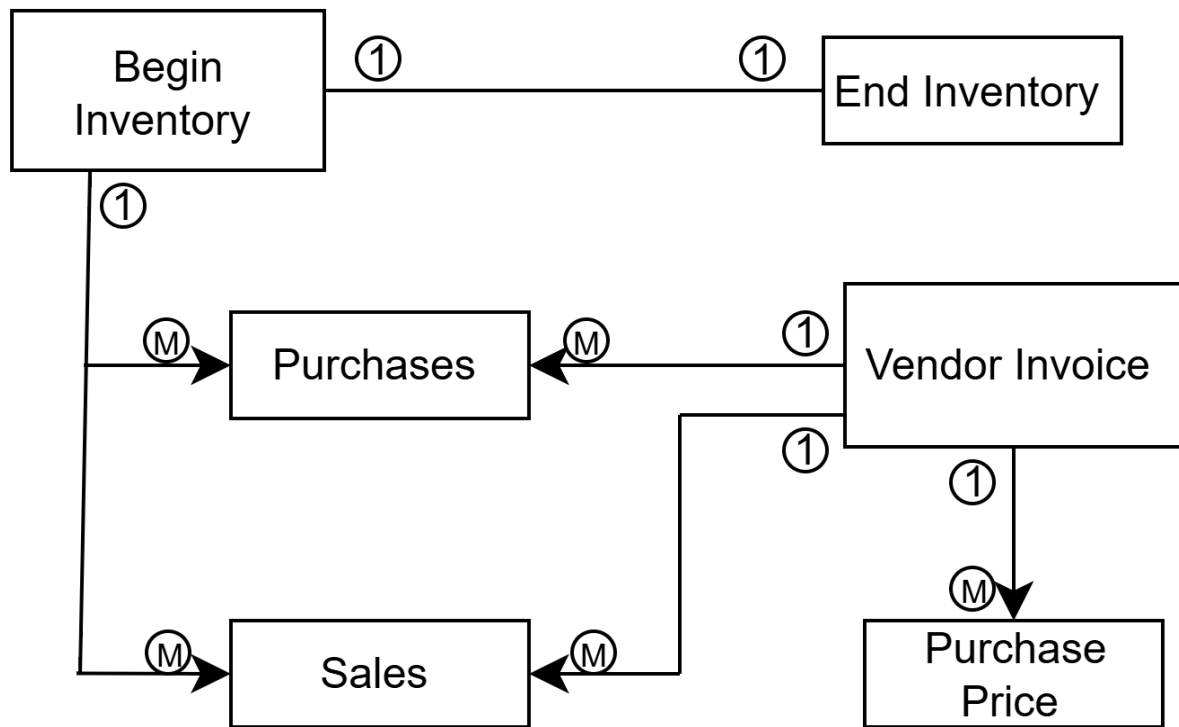
The system does not require high-end hardware and can run on a standard computer.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** Minimum 500 GB Hard Disk / SSD
- **Display:** Standard monitor with 1366×768 resolution or higher
- **Input Devices:** Keyboard and Mouse

DATA DICTIONARY

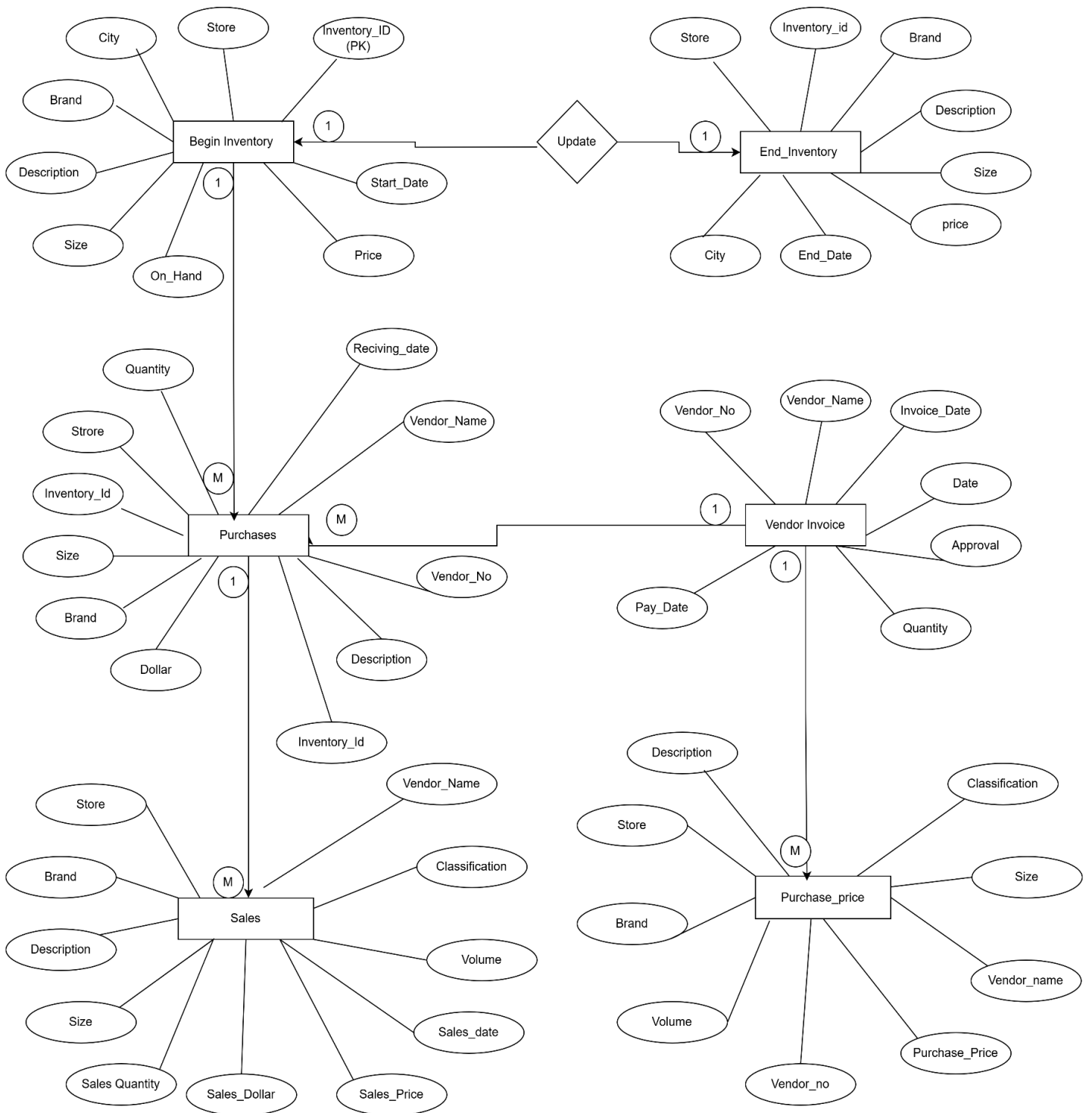
Field Name	Data Type	Description
VendorNumber	Integer	Unique identification number assigned to each vendor
VendorName	String	Name of the vendor supplying products
Brand	String	Brand name of the product
Description	String	Detailed description of the product
PurchasePrice	Float	Price at which the product was purchased from the vendor
ActualPrice	Float	Selling price of the product
TotalPurchaseQuantity	Integer	Total quantity of products purchased
TotalSalesQuantity	Integer	Total quantity of products sold
TotalPurchaseDollars	Float	Total amount spent on purchasing products
TotalSalesDollars	Float	Total revenue generated from sales
GrossProfit	Float	Profit obtained after subtracting purchase cost from sales revenue
ProfitMargin	Float	Percentage of profit earned from total sales
FreightCost	Float	Transportation or shipping cost for products
StockTurnover	Float	Rate at which inventory is sold and replaced over a period

RELATIONAL DATABASE DIAGRAM

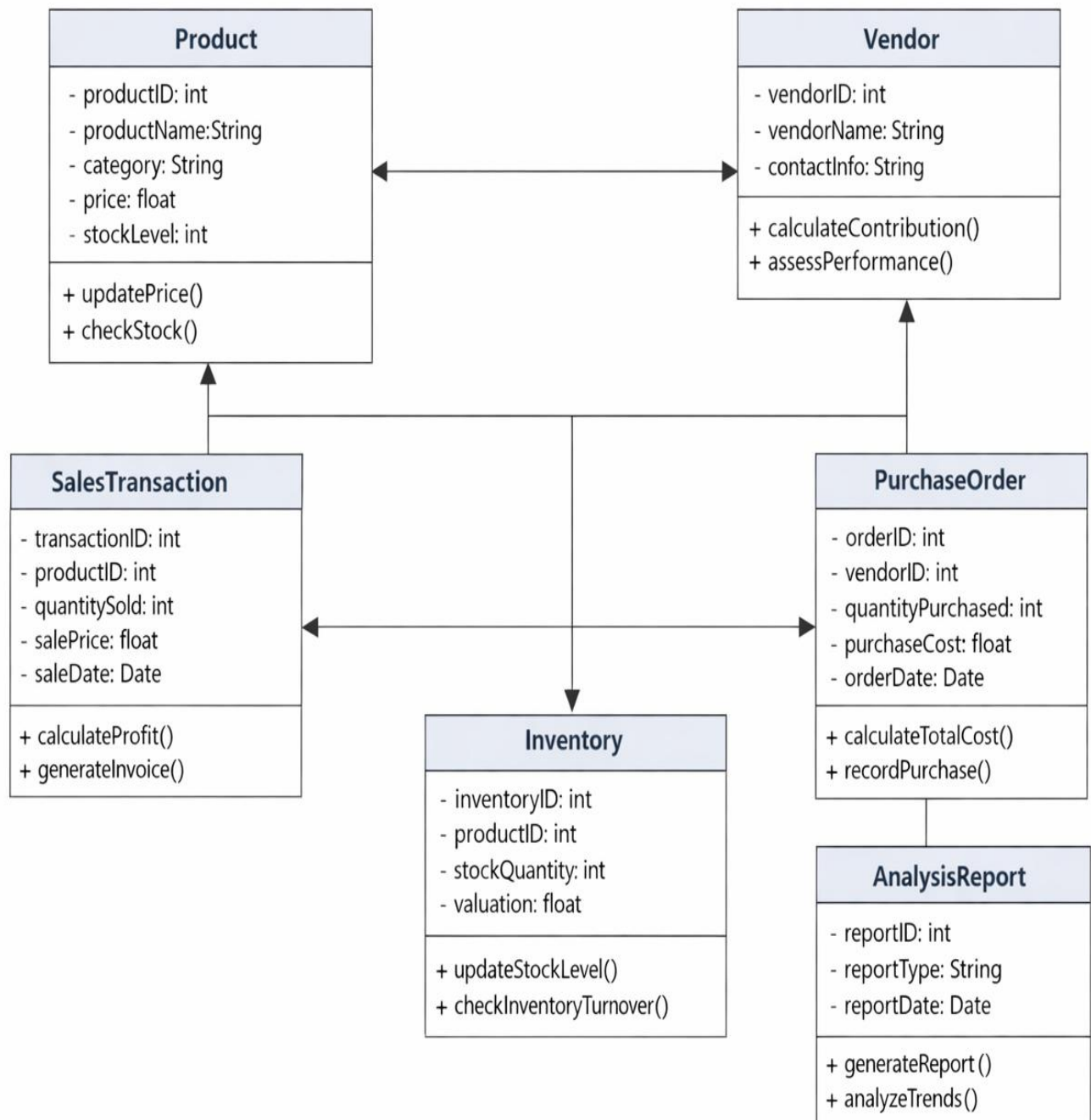


SYSTEM MODELING

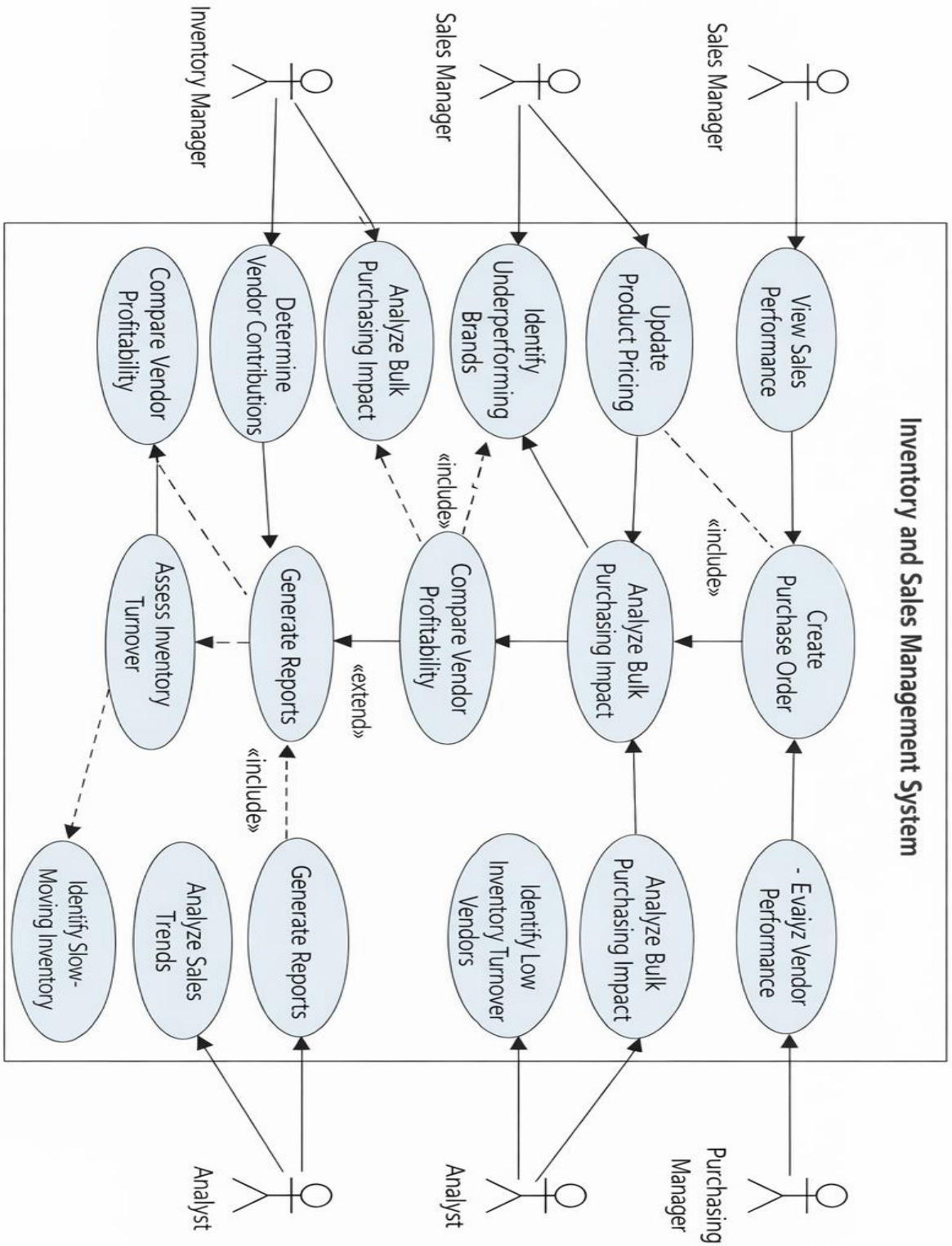
4.1 ER Diagram



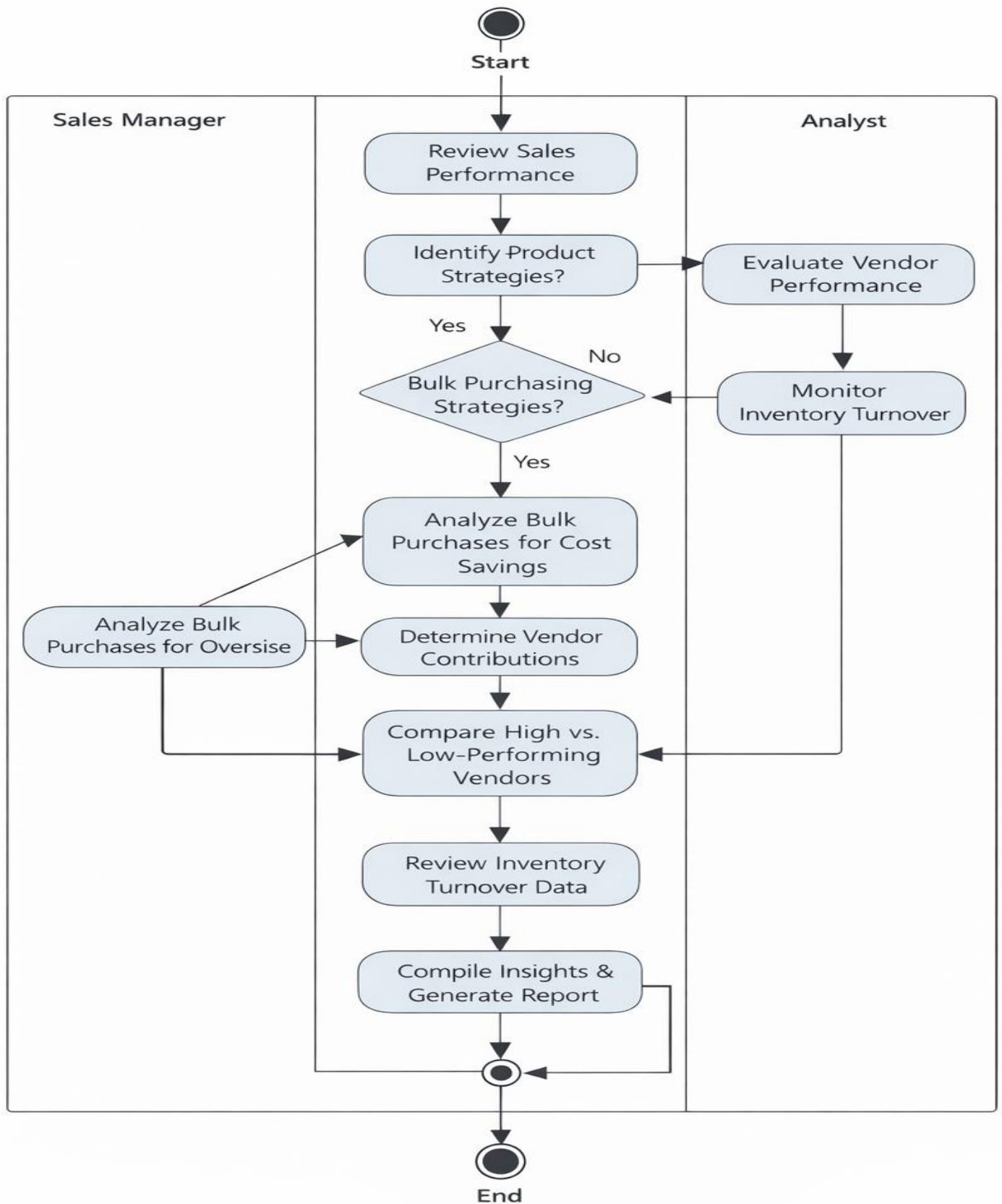
4.2 Class Diagram



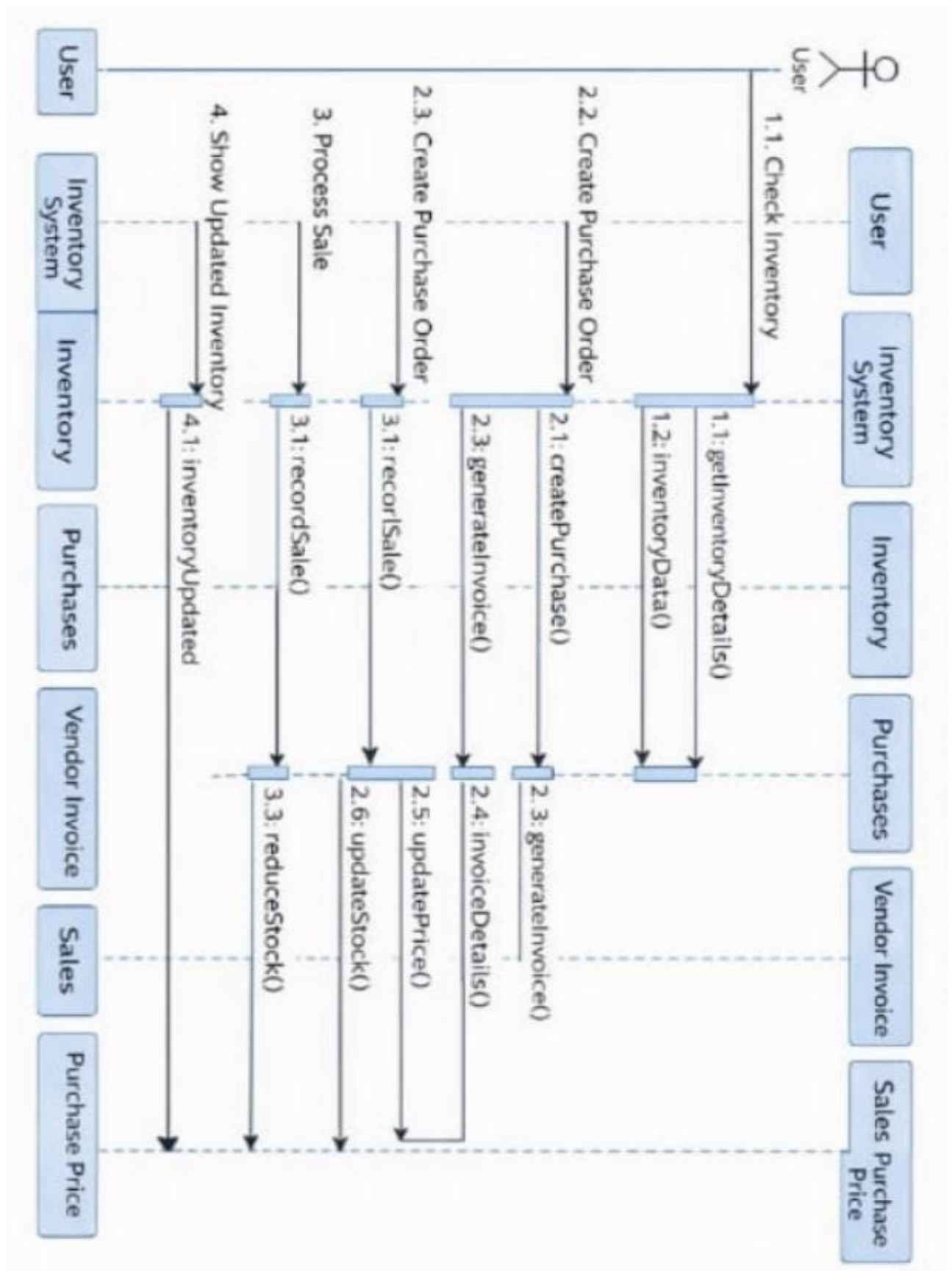
4.3 Use Case Diagram



4.4 Activity Diagram



4.5 Sequence Diagram

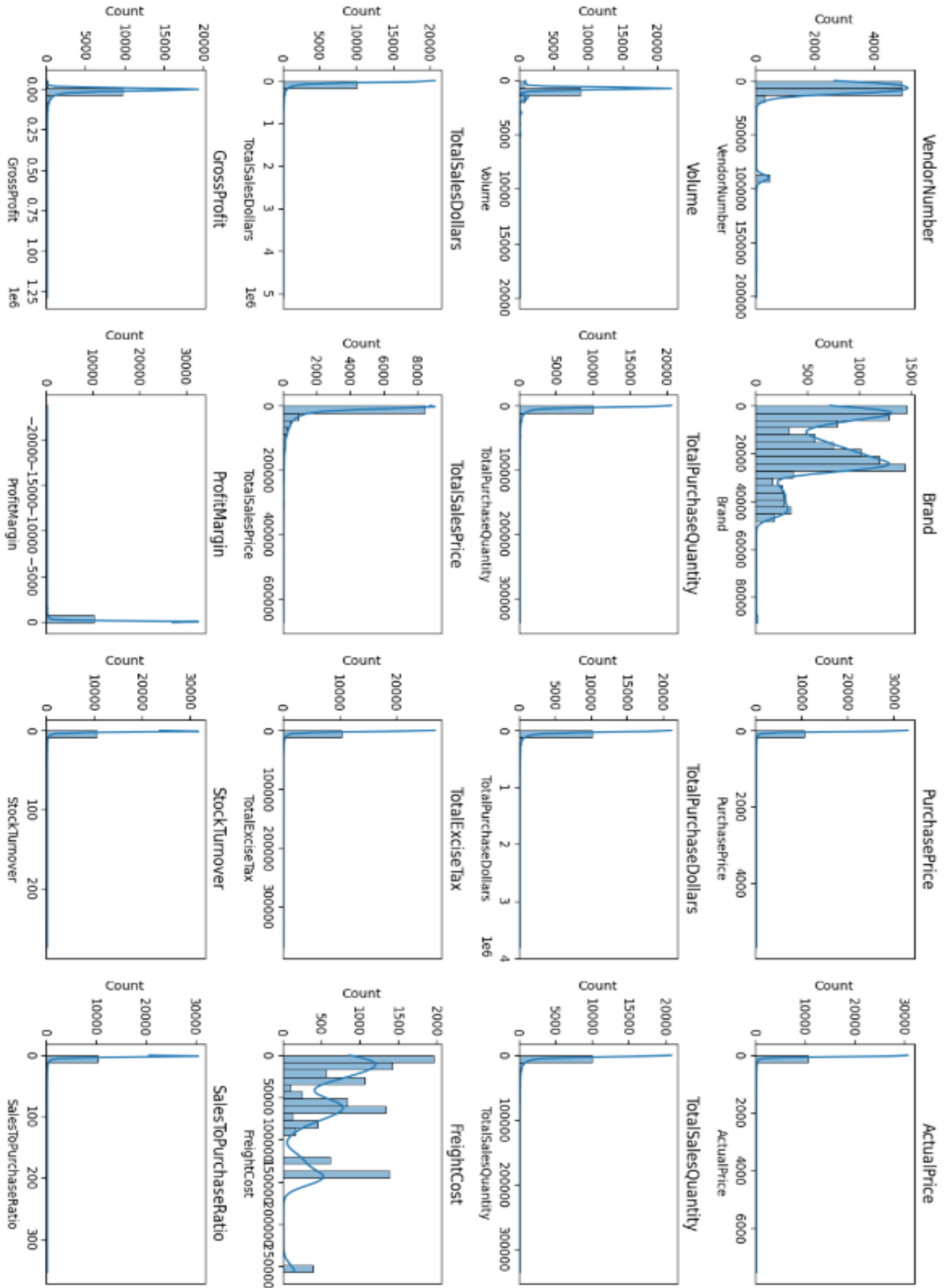


OUTPUT SCREENS

Exploratory Data Analysis Insights

Summary Statistics

1 #summary statistics										
2 df. describe().T										
	count	mean	std	min	25%	50%	75%	max		
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05		
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04		
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03		
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.390000	7.499990e+03		
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04		
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05		
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06		
TotalSalesQuantity	10692.0	3.077482e+03	10952.651391	0.00	33.000000	261.000000	1929.250000	3.349390e+05		
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06		
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05		
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05		
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05		
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06		
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01		
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02		
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02		



Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of -0, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

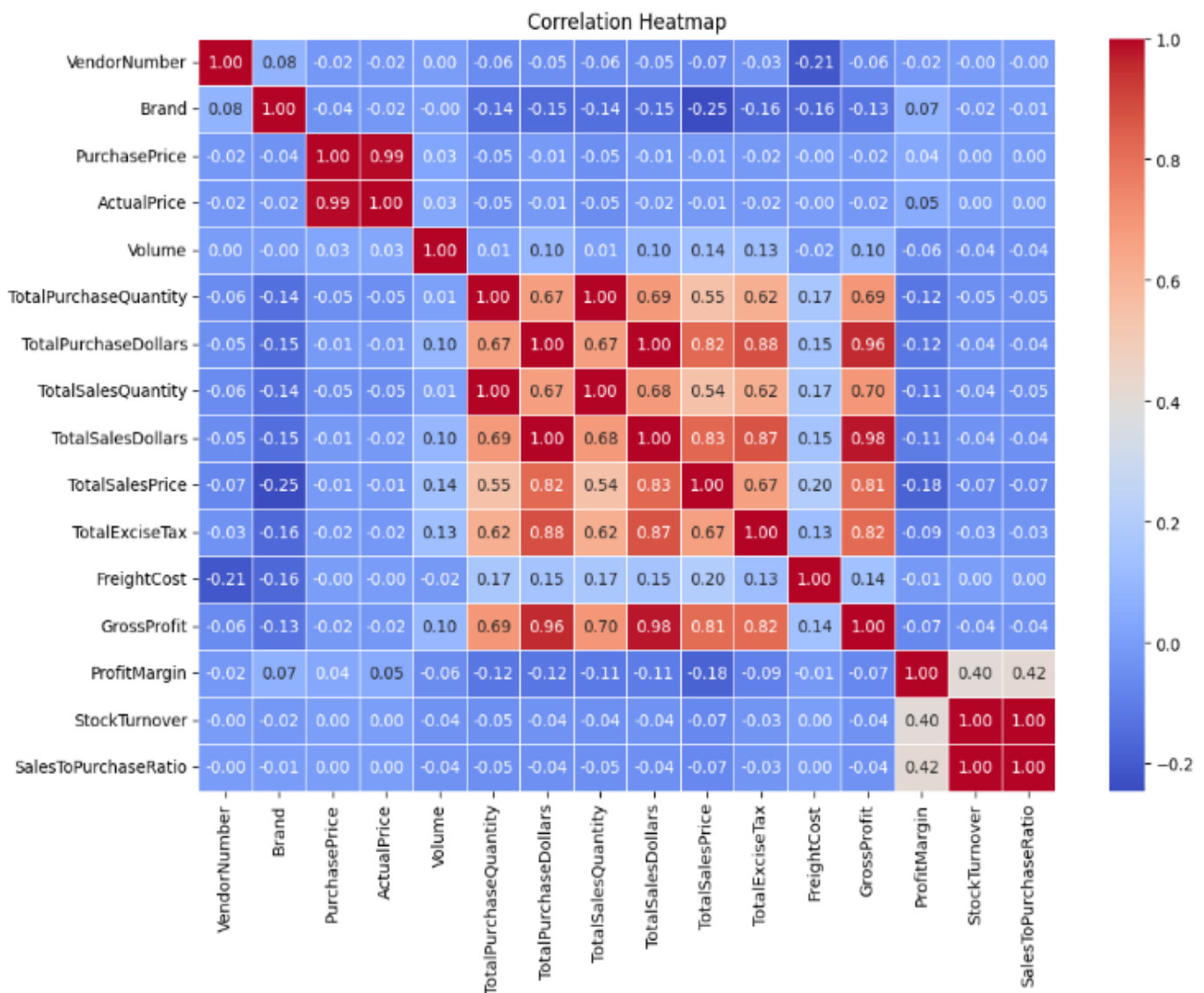
Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Questions & Key Findings

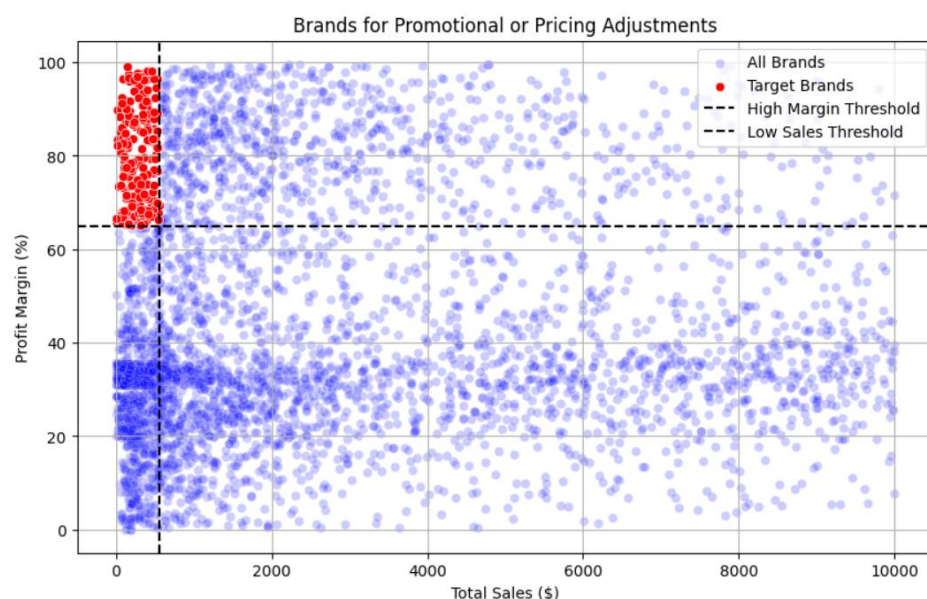
1. Brands for Promotional or Pricing Adjustments

Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournal 10	558.87	69.434752

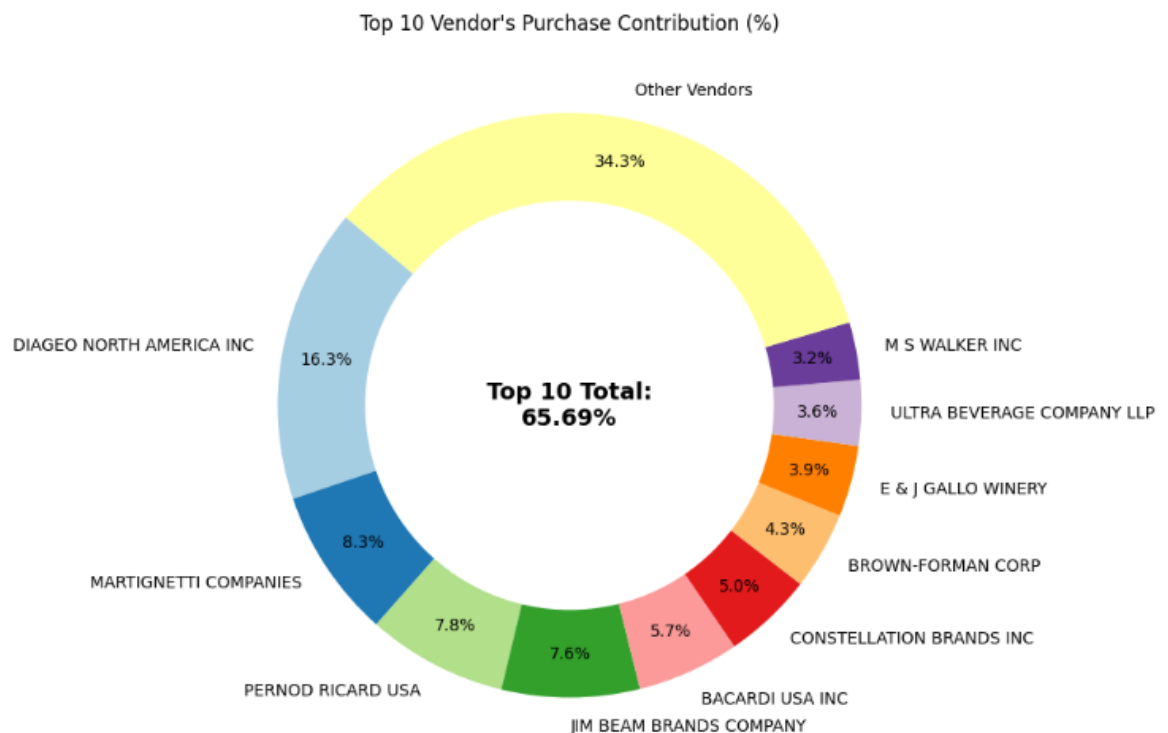
198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

UnitPurchasePrice	
OrderSize	
Small	39.068186
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: \$2.71M Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

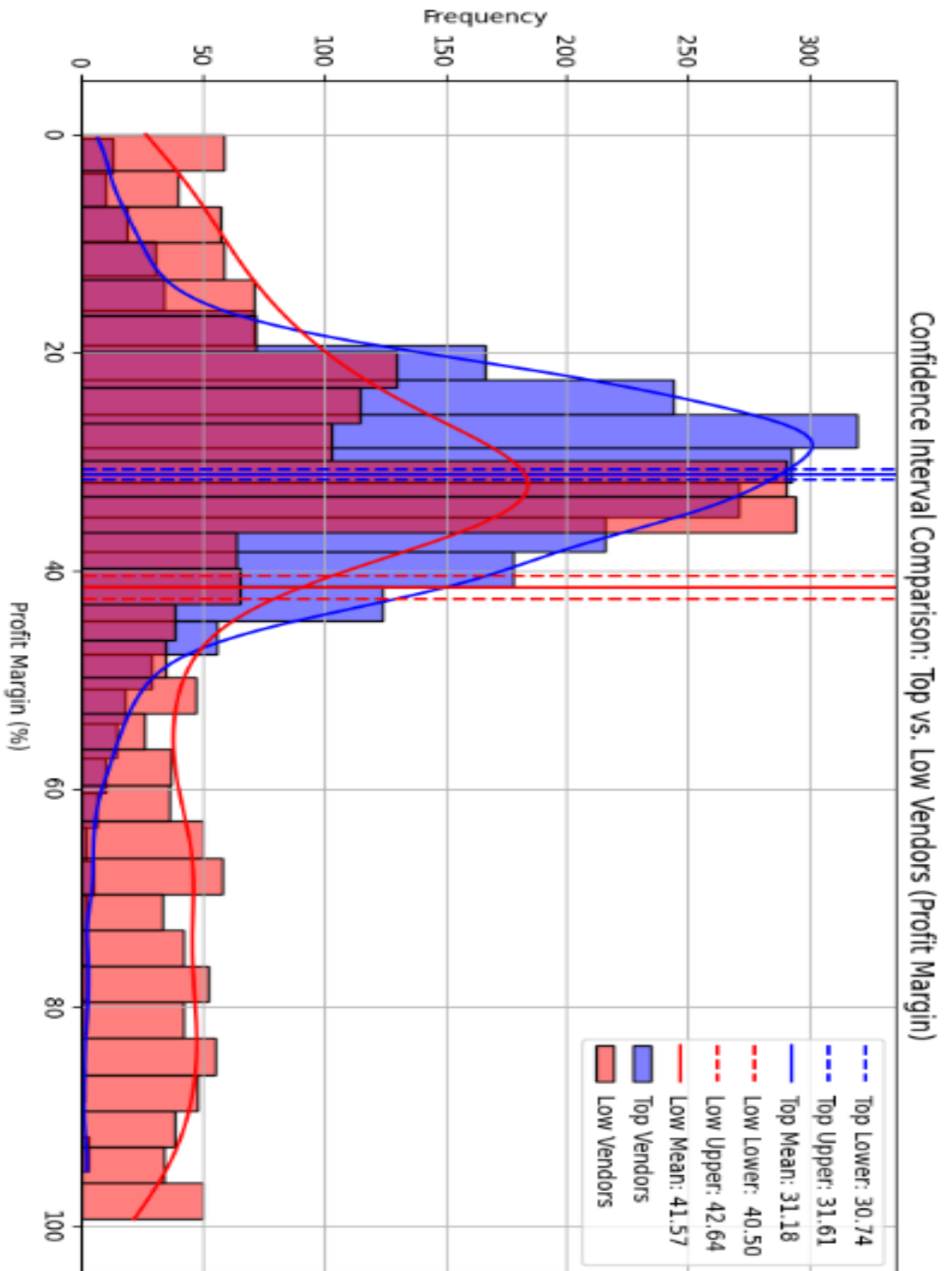
		VendorName	UnsoldInventoryValue
StockTurnover			
VendorName			
ALISA CARR BEVERAGES	0.615385	25 DIAGEO NORTH AMERICA INC	722.21K
HIGHLAND WINE MERCHANTS LLC	0.708333	46 JIM BEAM BRANDS COMPANY	554.67K
PARK STREET IMPORTS LLC	0.751306	68 PERNOD RICARD USA	470.63K
Circa Wines	0.755676	116 WILLIAM GRANT & SONS INC	401.96K
Dunn Wine Brokers	0.766022	30 E & J GALLO WINERY	228.28K
CENTEUR IMPORTS LLC	0.773953	79 SAZERAC CO INC	198.44K
SMOKY QUARTZ DISTILLERY LLC	0.783835	11 BROWN-FORMAN CORP	177.73K
TAMWORTH DISTILLING	0.797078	20 CONSTELLATION BRANDS INC	133.62K
THE IMPORTED GRAPE LLC	0.807569	61 MOET HENNESSY USA INC	126.48K
WALPOLE MTN VIEW WINERY	0.820548	77 REMY COINTREAU USA INC	118.60K

5. Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17% Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55% Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

Ho (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H1 (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

POWERBI DASHBOARD

Vendor Performance Dashboard

441.41M
Total Sales (\$)

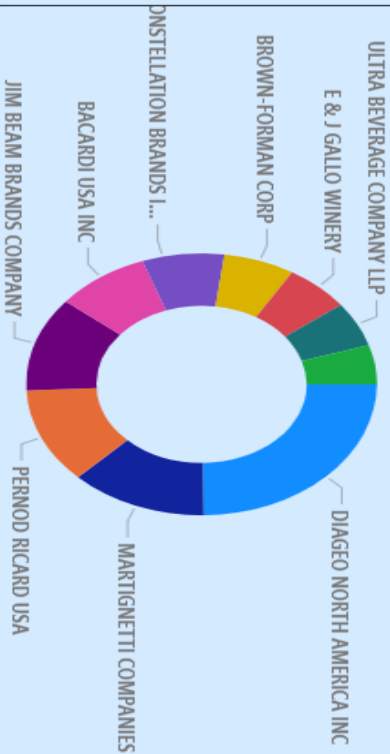
307.34M
Total Purchase (\$)

134.07M
Gross Profit (\$)

38.72
Profit Margin (%)

2.71M
Unsold Capital

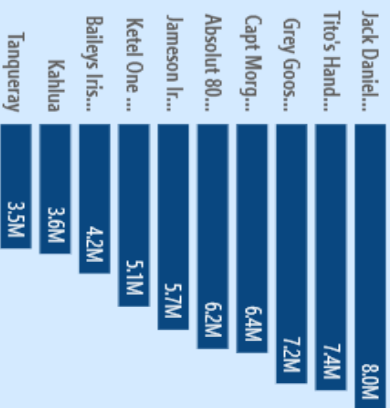
Purchase Contribution %



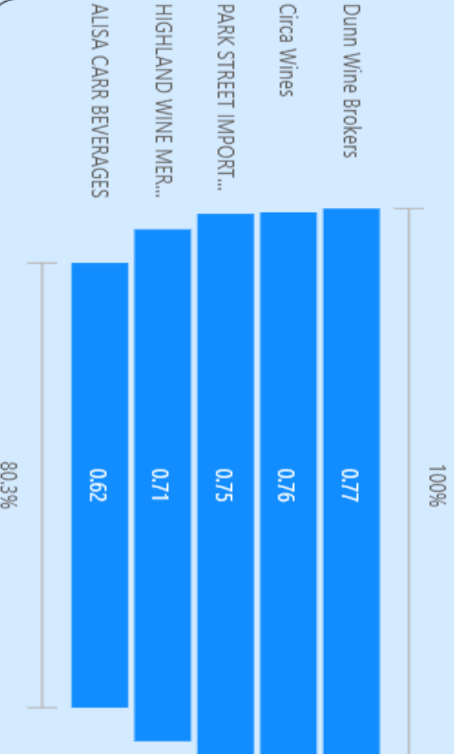
Top Vendors By Sales



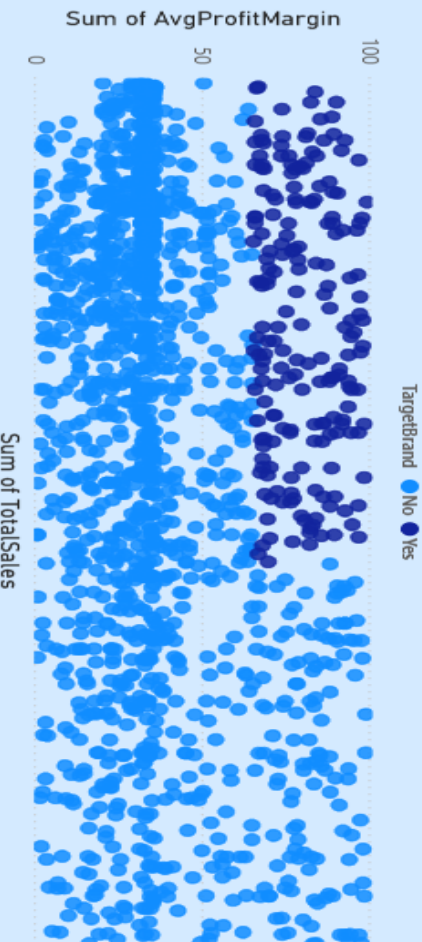
Top Brand By Sales



Low Performing vendors



Low Performing Brands

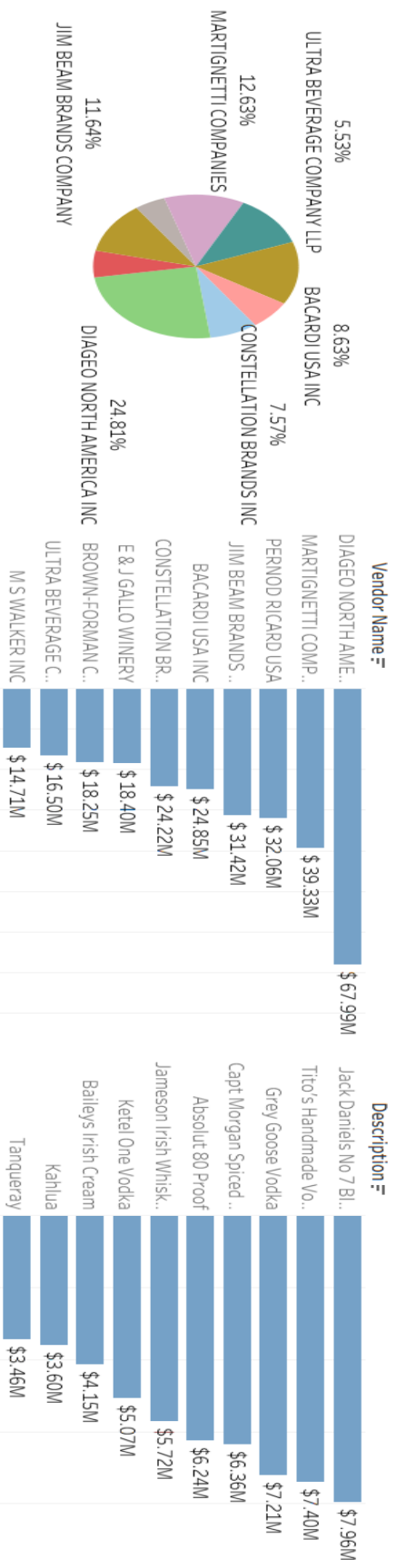


Vendor Performance Dashboard

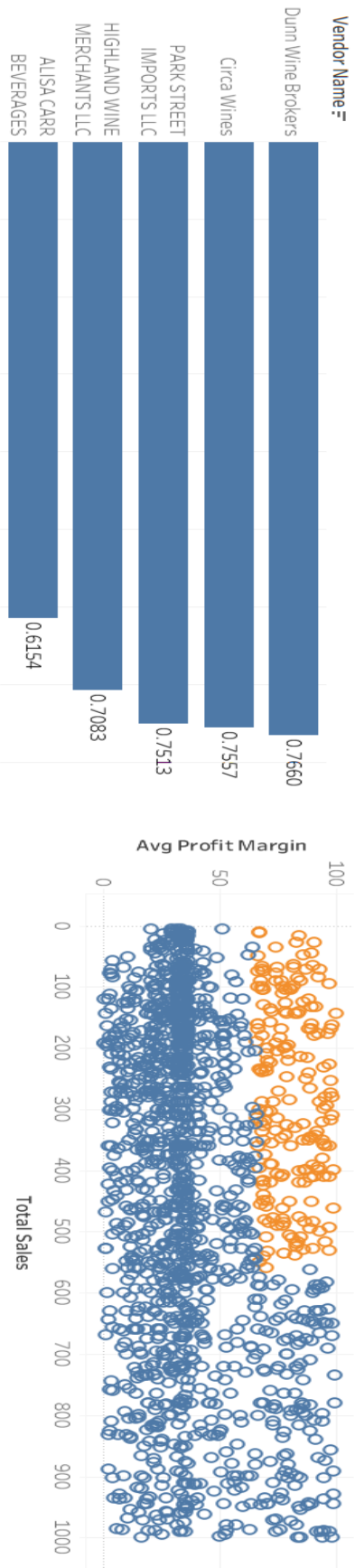
Unsold Capital

\$2.71M

Top Brand By Sales



Low Performing brands



Future scope

The project focuses on analysing sales, inventory, and vendor performance using historical data to generate insights that support business decision-making. While the system provides valuable analytical results, there are several opportunities to expand and improve the system in the future.

1. Real-Time Data Integration

In the future, the system can be enhanced by integrating real-time data from sales systems, inventory management systems, and enterprise databases. This would allow businesses to monitor sales performance, stock levels, and vendor activity instantly instead of relying only on historical datasets.

2. Machine Learning for Demand Forecasting

Advanced machine learning models can be incorporated to predict future product demand and sales trends. By forecasting demand accurately, organizations can optimize purchase quantities, reduce excess inventory, and avoid stock shortages.

3. Automated Inventory Optimization

The system can be upgraded to automatically recommend optimal inventory levels based on sales trends, seasonal demand, and supplier performance. This would help businesses reduce storage costs and improve stock availability.

4. Vendor Risk and Performance Monitoring

Future enhancements could include advanced vendor evaluation models that assess vendor reliability, delivery performance, pricing consistency, and risk factors. This would help organizations select the most reliable suppliers and reduce supply chain risks.

5. Advanced Data Visualization Dashboards

The system can be integrated with advanced business intelligence platforms to create interactive dashboards. These dashboards would allow managers to explore data visually, compare performance metrics, and make strategic decisions more effectively.

6. Integration with Enterprise Systems

The project can be extended by integrating with enterprise systems such as ERP (Enterprise Resource Planning) or supply chain management platforms. This integration would allow automated data flow between purchasing, sales, inventory, and financial systems.

7. Mobile and Web-Based Application

In the future, the system could be developed as a web-based or mobile application that allows managers to access reports, monitor inventory performance, and analyze vendor contributions from anywhere.

8. Automated Alerts and Recommendations

The system could be enhanced with automated alerts that notify managers when inventory levels are low, when products are not selling efficiently, or when vendor performance declines. Intelligent recommendation systems could also suggest pricing adjustments or promotional strategies.

9. Predictive Profitability Analysis

Future development can include predictive analytics to estimate future profit margins based on changes in pricing, demand, and supplier costs. This would help businesses make proactive decisions rather than reactive ones.

CONCLUSIONS

This project analysed vendor performance, sales trends, and inventory efficiency to understand the key factors influencing profitability in a retail and wholesale environment. By examining sales, purchases, profit margins, and vendor contributions, the project provided valuable insights that can help businesses improve their operational and financial performance.

The analysis shows that the organization generated **total sales of approximately \$441.41M**, while **total purchases accounted for \$307.34M**, resulting in a **gross profit of \$134.07M** and an overall **profit margin of 38.72%**. These figures indicate that the business is operating profitably; however, there are still opportunities to optimize vendor relationships and inventory management.

From the vendor contribution analysis, it is evident that a **small number of vendors contribute a significant portion of total sales and purchases**. Vendors such as **Diageo North America, Martignetti Companies, and Pernod Ricard USA** are among the top contributors to sales. While this indicates strong partnerships with these vendors, it also highlights a potential risk of **over-dependence on a limited group of suppliers**, which could impact the supply chain if disruptions occur.

The dashboard also highlights **low-performing vendors and brands**, which show lower sales contributions despite maintaining relatively higher profit margins. This suggests that these vendors may benefit from improved marketing strategies, distribution expansion, or pricing adjustments to increase sales volume.

Another important observation is the presence of **unsold inventory capital worth approximately \$2.71M**, indicating that some products remain in stock without generating revenue. Slow-moving inventory can increase storage costs and reduce cash flow efficiency. Therefore, businesses should focus on optimizing purchase quantities and promoting slow-moving products through discounts or targeted campaigns.

The analysis of **brand performance and vendor profitability** also shows that while some brands generate high sales volumes, others maintain higher margins but lower demand. Balancing these two factors is essential for maximizing both revenue and profitability.

Overall, this project demonstrates how **data-driven analysis can support strategic decision-making in inventory management, vendor evaluation, and pricing strategies**. The insights generated from the dashboard enable organizations to identify underperforming vendors, optimize purchasing strategies, reduce inventory inefficiencies, and strengthen profitable vendor relationships.

In conclusion, implementing data analytics in business operations can significantly improve efficiency, reduce risks, and enhance profitability. The findings of this project provide actionable insights that can help organizations achieve better inventory control, stronger vendor management, and sustainable business growth.

BIBLIOGRAPHY

Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques*. Morgan Kaufmann Publishers.

This book provides concepts and techniques related to data mining and data analysis used for extracting useful insights from large datasets.

Provost, F., & Fawcett, T. (2013). *Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking*. O'Reilly Media.

This reference explains how data analytics and business intelligence can support better decision-making in organizations.

Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modelling*. Wiley Publications.

This book helped in understanding how business data can be organized and analyzed for reporting and analytics.

Dataset: https://topmate.io/ayushi_mishra/1557424